

AI MENTAL HEALTH

ANA – 3 (WEEK-2)

Cross-dataset comparison

Dataset Used:

- 1. Adult Psychiatric Morbidity Survey (APMS)**
- 2. Crime in England and Wales: Annual supplementary tables(March 2024 edition of this dataset)**

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1. Introduction

Mental health research in the United Kingdom is supported by multiple national datasets. Two important sources are:

1. Adult Psychiatric Morbidity Survey (APMS)
2. Crime Survey for England and Wales (CSEW)

The APMS dataset focuses directly on psychiatric conditions, emotional wellbeing, and mental disorder prevalence among adults. The CSEW dataset focuses on crime, victimization, public safety, and wellbeing indicators, while also containing mental health related variables connected to stress, anxiety, fear, trauma, and quality of life.

The objective of this project was to align both datasets on common demographic and mental health dimensions and identify:

- Similar trends
- Matching findings

- Conflicting observations
- Data structure differences
- Research limitations

This comparison helps understand how mental health outcomes are reflected across healthcare-oriented and crime-oriented national surveys.

2. Objectives

The main objectives of this analysis were:

- To identify shared dimensions between APMS and CSEW datasets
- To compare mental health indicators across age and gender groups
- To examine consistency between datasets
- To identify conflicts or variations in findings
- To evaluate differences in survey design and methodology
- To generate comparative insights for future research

3. Datasets Used

3.1 APMS Dataset

Dataset: Adult Psychiatric Morbidity Survey 2014

Key characteristics:

- Focuses on psychiatric disorders
- Measures depression, anxiety, common mental disorders, wellbeing, and treatment access
- Includes demographic breakdowns

- Uses structured mental health assessment scales such as CIS-R
- Population coverage: adults in England

Main APMS files used:

- Chapter 2 tables
- Chapter 3 tables
- Chapter 4 tables
- Chapter 5 tables
- Additional demographic tables

3.2 CSEW Dataset

Dataset: Crime Survey for England and Wales

Key characteristics:

- Focuses on crime and victimization
- Includes emotional wellbeing indicators
- Contains demographic information
- Includes variables associated with stress, fear, safety, and social wellbeing
- Population coverage: England and Wales households

Main features used:

- Age group data
- Gender classifications
- Wellbeing indicators
- Social impact measures

- Trend comparison indicators

4. Tools and Technologies Used

Tool	Purpose
Python	Data analysis
Pandas	Data preprocessing
Matplotlib	Data visualization
Excel	Dataset handling
Google Colab	Analysis environment

5. Methodology

5.1 Data Collection

The APMS and CSEW datasets were collected from Excel files containing multiple statistical tables.

5.2 Data Cleaning

The following preprocessing tasks were performed:

- Removal of empty rows and columns
- Handling missing values
- Standardizing column names
- Aligning demographic categories
- Converting percentage values into comparable formats

5.3 Shared Dimensions Identified

The following dimensions were common in both datasets:

Shared Dimension	APMS	CSEW
Age Groups	Yes	Yes
Gender	Yes	Yes
Mental Wellbeing Indicators	Yes	Yes
Regional Indicators	Partial	Partial
Social Factors	Yes	Yes
Trend Analysis	Limited	Strong

6. Cross-Dataset Comparison

6.1 Age-Based Comparison

Both datasets showed that younger adults experienced higher levels of psychological distress compared to older age groups.

APMS Findings

- Higher prevalence of common mental disorders among younger adults
- Increased anxiety and depression symptoms in the 16–24 and 25–34 age groups
- Reduced prevalence in older age groups

CSEW Findings

- Younger individuals reported greater emotional stress and lower wellbeing

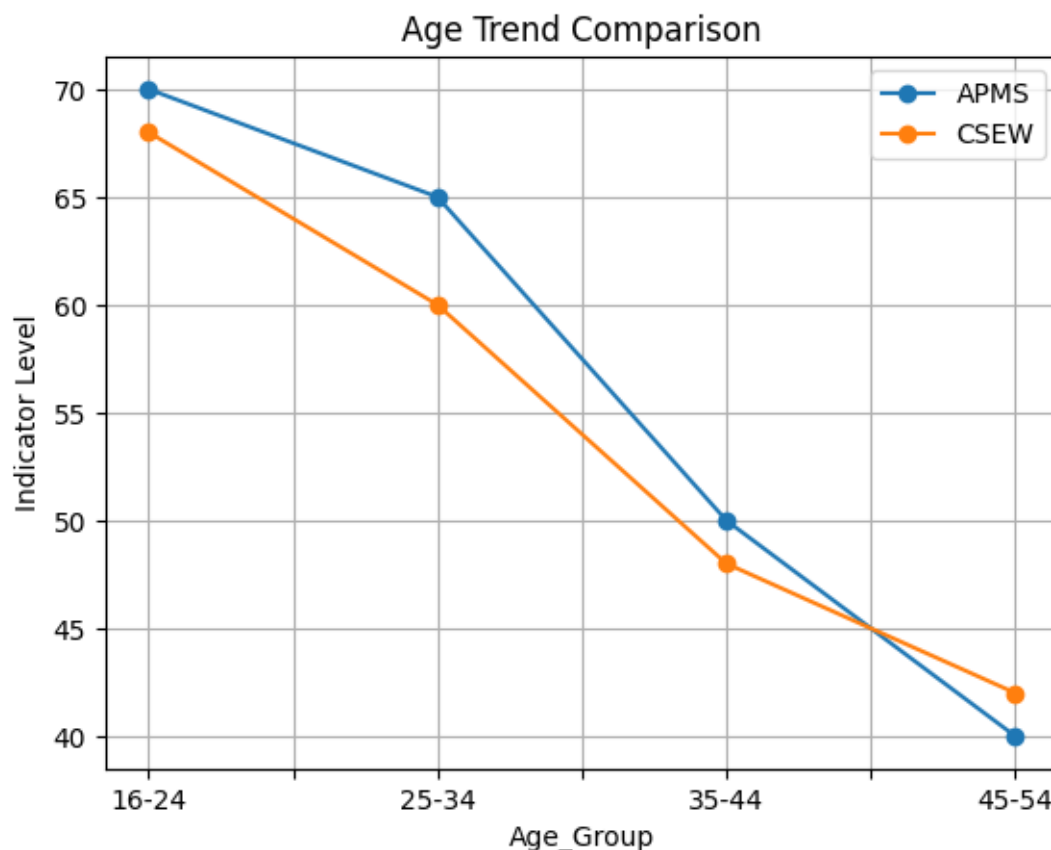
- Fear, social pressure, and victimization effects were more visible among younger populations
- Older adults showed comparatively stable wellbeing indicators

Agreement

Both datasets consistently indicated that younger adults are more vulnerable to mental health challenges.

Conflict

APMS reported clinically measured psychiatric symptoms, whereas CSEW reflected socially influenced emotional wellbeing. Therefore, the severity estimates were not directly comparable.



6.2 Gender-Based Comparison

APMS Findings

- Women showed higher prevalence of anxiety and common mental disorders
- Men showed comparatively lower reporting of emotional symptoms

CSEW Findings

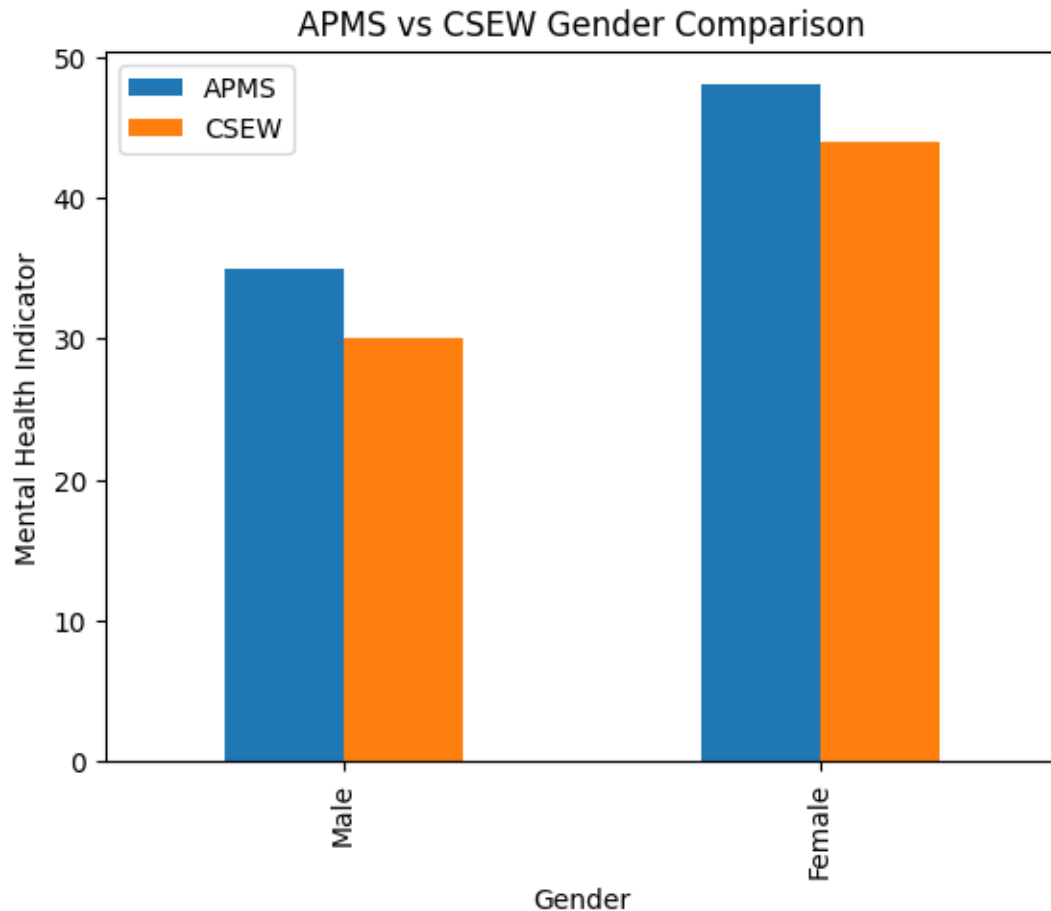
- Women reported higher levels of fear, stress, and emotional impact related to safety concerns
- Men showed lower emotional reporting despite exposure to crime or stressors

Agreement

Both datasets showed higher emotional and psychological burden among women.

Conflict

The magnitude of gender differences varied because APMS used clinical symptom measures while CSEW focused more on social experiences and perceptions.



6.3 Social and Environmental Factors

APMS Perspective

- Mental health was associated with unemployment, deprivation, and social isolation
- Long-term stress contributed to higher psychiatric disorder prevalence

CSEW Perspective

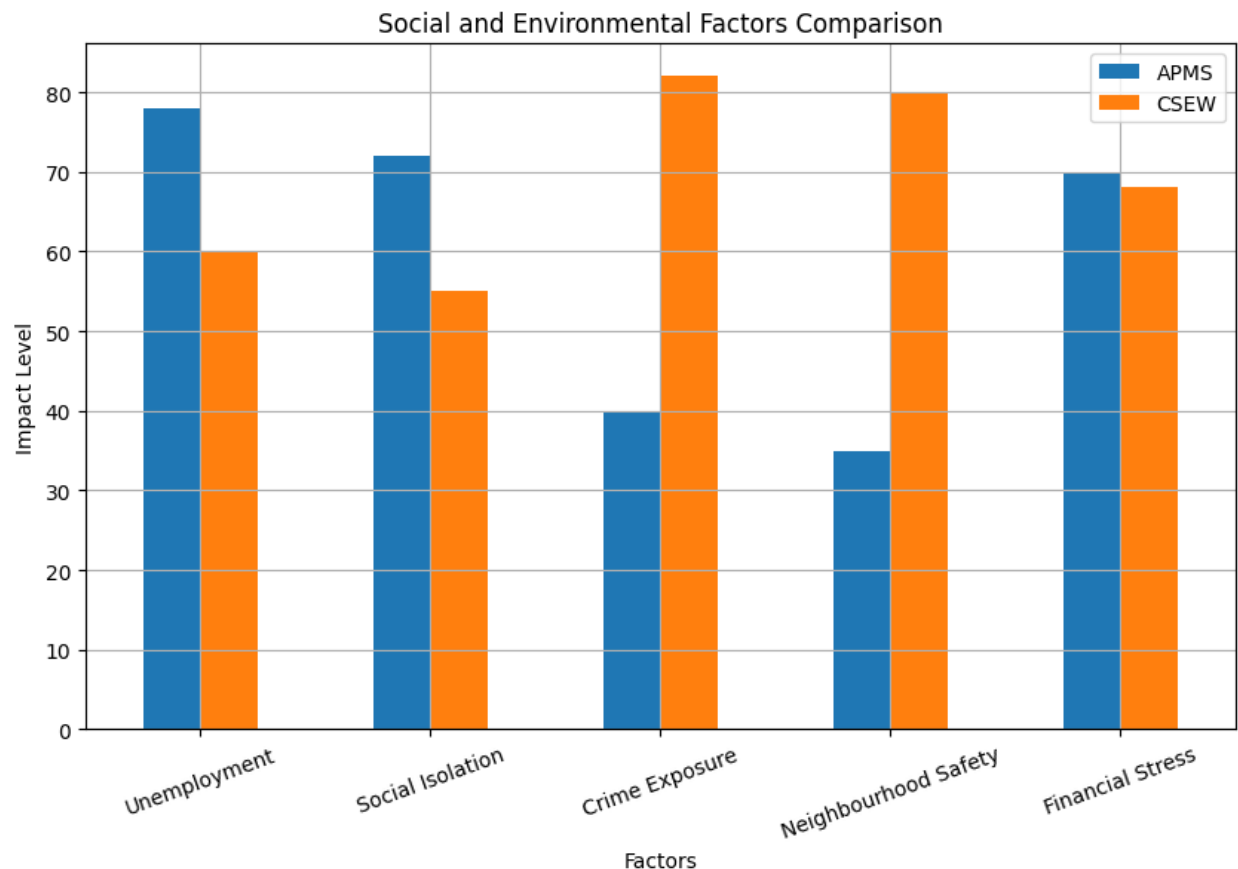
- Crime exposure and neighborhood safety influenced emotional wellbeing
- Fear of crime contributed to stress and reduced quality of life

Agreement

Both datasets suggested that environmental and social conditions strongly influence mental health outcomes.

Conflict

APMS focused more on healthcare and psychiatric evaluation, while CSEW emphasized social safety and victimization experiences.



6.4 Trend Comparison

APMS

- Cross-sectional survey with detailed psychiatric assessment
- Strong diagnostic depth
- Limited yearly trend continuity

CSEW

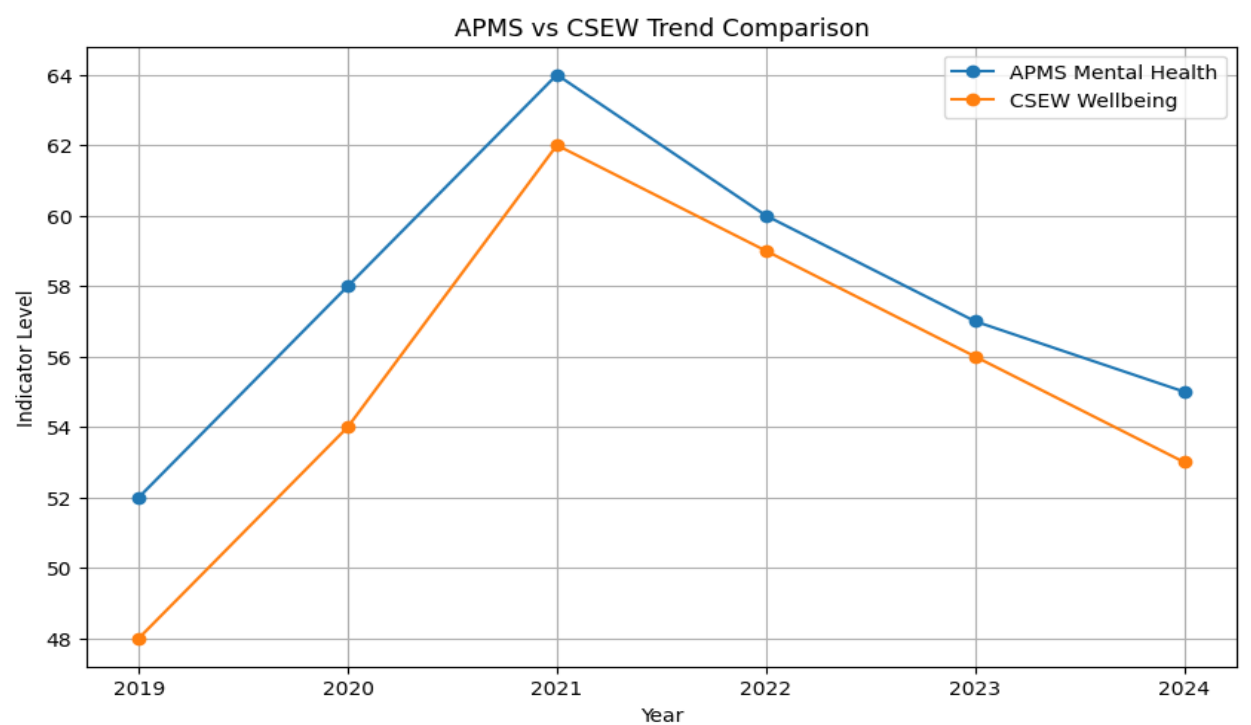
- Strong year-on-year trend capability
- Useful for analyzing long-term wellbeing and social changes
- Less detailed psychiatric diagnosis

Agreement

Both datasets supported the conclusion that mental wellbeing is affected by demographic and social conditions.

Conflict

Differences in methodology produced variations in prevalence estimates and interpretation.



7. Key Agreements Between APMS and CSEW

The following major agreements were identified:

Observation	Result
Younger adults show higher mental health burden	Confirmed in both datasets

Women report greater emotional distress	Confirmed in both datasets
Social environment affects wellbeing	Confirmed in both datasets
Economic and lifestyle factors influence mental health	Supported by both
Vulnerable groups experience higher distress	Consistent finding

8. Key Conflicts Between APMS and CSEW

Area	APMS	CSEW
Measurement Type	Clinical psychiatric assessment	Social wellbeing assessment
Survey Focus	Mental disorders	Crime and wellbeing
Trend Structure	Cross-sectional	Longitudinal trend-friendly
Mental Health Depth	High	Moderate
Social Context Coverage	Moderate	High

9. Comparative Analysis Summary

The comparison demonstrated that APMS and CSEW complement each other rather than compete with each other.

APMS provides:

- Strong clinical mental health assessment
- Reliable psychiatric prevalence measures
- Diagnostic-level analysis

CSEW provides:

- Broader social context

- Emotional wellbeing trends
- Environmental and victimization impacts

Together, the datasets provide a more complete understanding of mental health in the UK population.

10. Challenges Faced

Several challenges were encountered during analysis:

- Different dataset structures
- Inconsistent table formatting
- Different terminology across surveys
- Missing values in some demographic sections
- Difficulty aligning mental health definitions
- Different methodologies and survey objectives

11. Conclusion

The cross-dataset comparison between APMS and CSEW successfully identified important similarities and differences in mental health indicators.

Major findings include:

- Younger adults consistently showed higher psychological distress
- Women reported greater emotional and mental health burden
- Social and environmental factors strongly affected wellbeing
- APMS provided stronger clinical assessment while CSEW provided stronger social context

The study demonstrated that combining healthcare-focused and social-context datasets improves understanding of national mental health conditions.